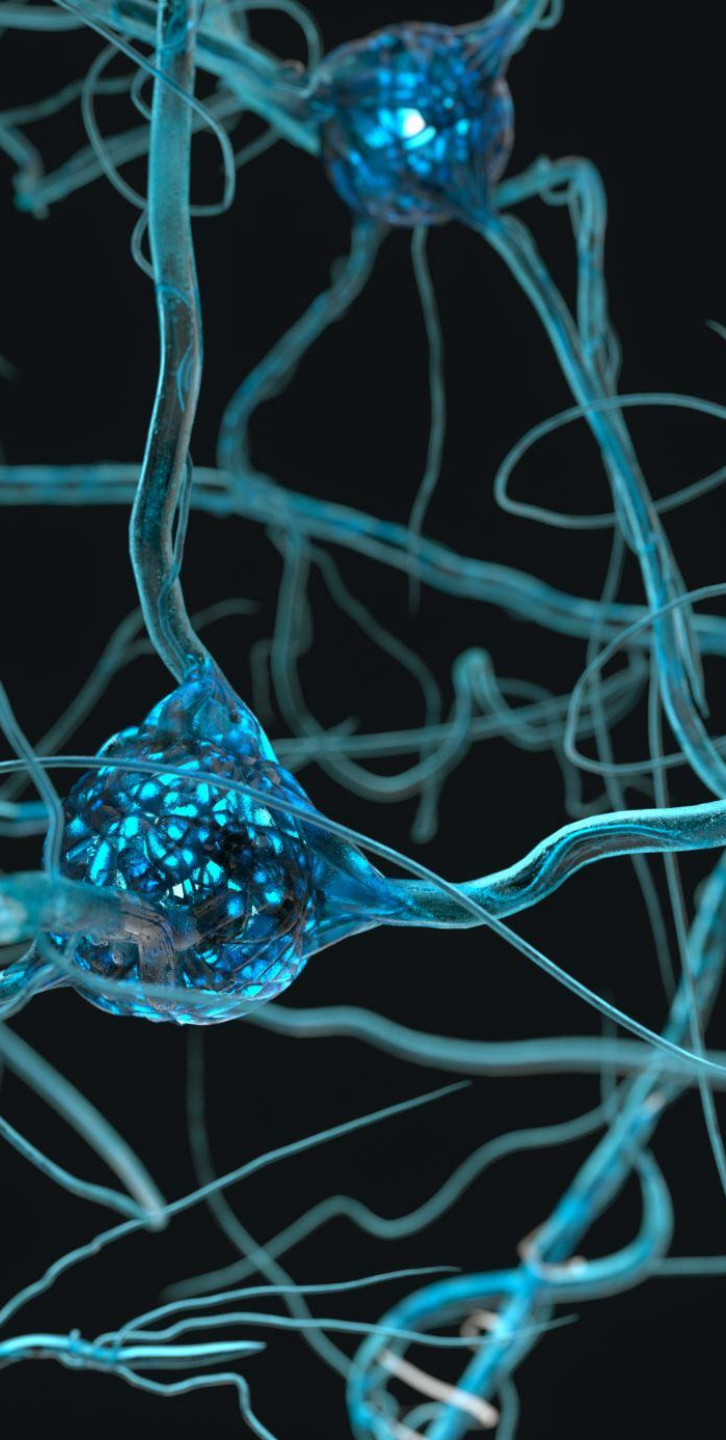




# Molecular Tumor Board: Current Role of Variant Interpretation and the Future with OakVar

Where we are, where we're going, and  
what it will take to get there

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# Molecular Tumor Board

- Goals: interpret molecular data, guide precision therapy, ensure clinical relevance

# Where We Are Now

- Current state: manual effort, disparate tools, fragmented data integration
- Achievements: standardized pipelines emerging (CAN.HEAL), interoperability initiatives (FHIR Genomics)

**1. A decalogue of Molecular Tumor Board (MTB)  
recommendations from the CAN.HEAL Consortium**  
Ciliberto, Gennaro et al.  
European Journal of Cancer, Volume 222, 115433

# Ciliberto, Gennaro et al. provide great context

- A Virtual MTB session platform with Decision-Support Tool (DST)-embedded logistics should have these features

|                                      |                                   |                                          |                                                              |                                            |                                                                         |
|--------------------------------------|-----------------------------------|------------------------------------------|--------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------|
| Controlled data access               | Session scheduling                | WEB conferencing                         | <b>Graphical web interface</b>                               | Graphical timeline of the decision process | <b>Annotation empowering field-specific knowledge</b>                   |
| <b>Customizable content</b>          | External data input               | Link to external drug-matching databases | A clinical trial matching system                             | <b>Reference to sources</b>                | <b>Adherence to a European Electronic Health Record Exchange Format</b> |
| <b>A format for easy data export</b> | Collection of MTB recommendations | <b>Clinical outcome collection</b>       | Artificial intelligence (AI) and machine learning (ML) tools | Cybersecurity                              |                                                                         |

\*table created by Akash using information from the study

# Annotations and Their Place in the Board

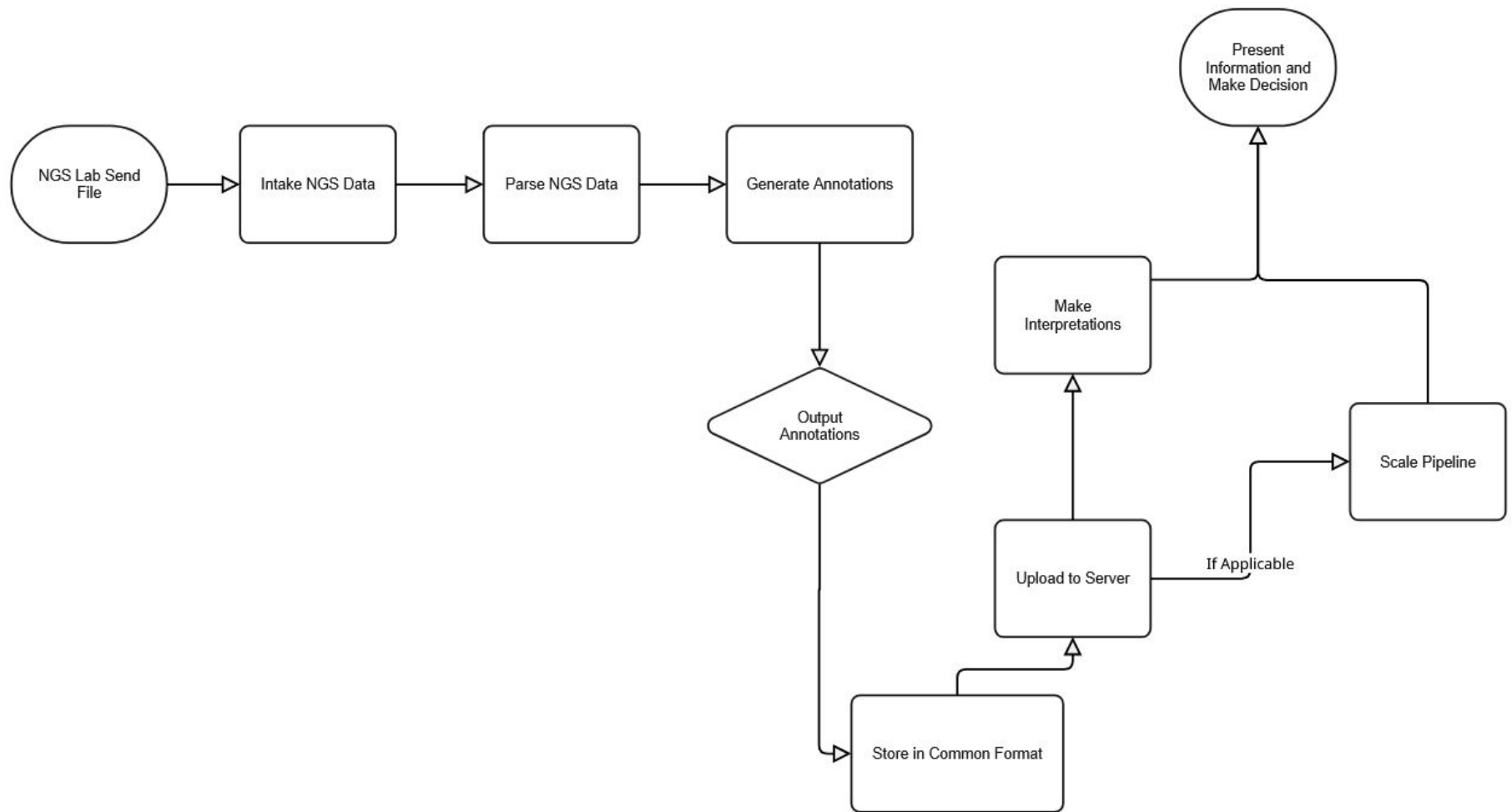
- Variant Types: somatic mutations, CNVs, fusions, germline variants
- Role: targeted therapies, clinical trial eligibility, prognostic insights
- Challenges: volume of data, conflicting annotations, reproducibility

# Where We Are Going

- Seamless integration of variant data into clinical systems
- AI-assisted interpretation and prioritization
- Global standards adoption (FHIR, GA4GH, ClinGen)
- Scalable, cloud-native MTB platforms (possibly)

# What It Will Take to Get There

- Technology: scalable pipelines, cloud, interoperability standards
- Collaboration: clinicians, informaticians, policymakers
- Data Infrastructure: harmonized databases, shared knowledgebases
- Empowering MTBs with key bioinformatic metadata (annotations)





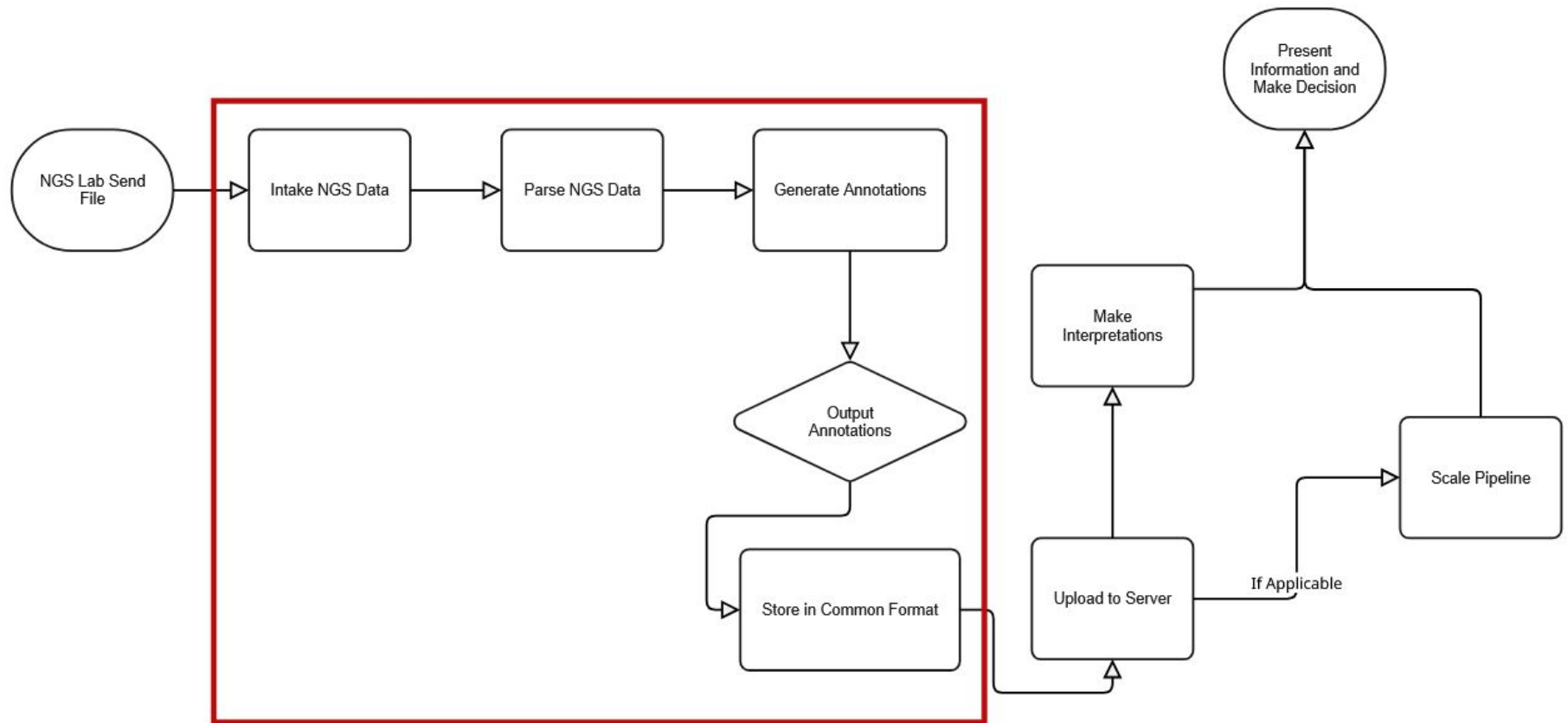
# OakVar as a Facilitator

- Genomic variant interpretation platform
- Benefits: standardized pipelines, modular annotation, scalability
- Example: automating annotation for board-ready reports



OakVar

# Steps that OakVar performs



# The Golden Ticket!

## Nextflow file detailing

- OakVar Ingestion
- Multiple Annotations in one step
- Scalable onto cloud technology!

```
#!/usr/bin/env nextflow
```

```
params.vcf          = "s3://your-bucket/input/example_patient_0254.vcf"
params.outdir       = "s3://your-bucket/output/"
params.modules      = "gnomad3 cancerhotspots oncoKB chasmpus"
```

```
process annotateVariants {
    executor 'local'
    container 'oakbioinformatics/oakvar:2.8.1'

    input:
    path vcf_file from Channel.fromPath(params.vcf)
```

```
    output:
    path "results/*" into annotated_results
```

```
    script:
    """
    mkdir -p results
    ov run \
        --input \${vcf_file} \
        --output-dir results \
        --modules \${params.modules} \
        --skip-report
    """
```

```
}
```

```
process uploadResults {
    executor 'local'
```

```
    input:
    path annotated_dir from annotated_results
```

```
    script:
    """
    aws s3 cp --recursive \${annotated_dir} \${params.outdir}
    """
```

```
}
```

```

{
  "fullUrl": "urn:uuid:fd2210fa-e15d-3d18-301d-22c6e8334c4e",
  "resource": {
    "resourceType": "Observation",
    "meta": {
      "profile": [
        "https://build.fhir.org/ig/HL7/genomics-reporting/StructureDefinition-genomic-annotation.html"
      ]
    },
    "status": "final",
    "code": {
      "coding": [
        {
          "system": "http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/tbd-codes-cs",
          "code": "genomic-annotation"
        }
      ]
    },
    "subject": {
      "reference": "urn:uuid:a9ef8dce-df28-d438-00ac-b15f6083a846",
      "type": "Reference"
    },
    "component": [
      {
        "code": {
          "coding": [
            {
              "system": "http://hl7.org/fhir/uv/genomics-reporting/CodeSystem/genomic-annotation-pipeline",
              "code": "annotation-pipeline",
              "display": "Annotation Pipeline"
            }
          ]
        },
        "valueString": "OakVar: 2.12.21"
      },
      {
        "code": {
          "coding": [
            {
              "system": "http://example.org",
              "code": "23589-5",
              "display": "Highest Sensitive Level"
            }
          ]
        },
        "valueString": "Level 2 – clinical evidence"
      }
    ]
  }
}

```

## Native FHIR Output

- Intuitive yml-based module allowing for devs to write different implementations
- Collaboration is needed for future refinement and

# Collaboration and Discussion is Key!

- Data modelling is a sisyphean feat
- Potential points of discussion
  - Representative Test Data (example files)
  - Accurate Cloud Infrastructure (how the files are stored, how can they be accessed?)
  - Production level requirements

<https://docs.google.com/spreadsheets/d/1RGfjxPhFB0HmAAIUOmmzW6YT1br2C0F8808XhsNnQ3E/edit?usp=sharing>





# Key Takeaway

Variant Annotations help bring actionable insights to the data produced.



Thank you very much!